PRACTICAL NO 3: CAMERA CALIBRATION

```
import numpy as np
import cv2
import glob

# Define the number of corners in the chessboard
num_corners_x = 9
num_corners_y = 6

# Prepare object points, like (0,0,0), (1,0,0), (2,0,0) ....,(6,5,0)
objp = np.zeros((num_corners_x * num_corners_y, 3), np.float32)
objp[:, :2] = np.mgrid[0:num_corners_x, 0:num_corners_y].T.reshape(-1, 2)

# Arrays to store object points and image points from all the images
objpoints = []  # 3d point in real world space
imgpoints = []  # 2d points in image plane.

# Load images
images = glob.glob("D:\CV\calibration_images\*.jpg")

# Loop through images and find chessboard corners
for fname in images:
    img = cv2.imread(fname)
    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

    # Find the chessboard corners
    ret, corners = cv2.findChessboardCorners(gray, (num_corners_x, num_corners_y), None)

    # If found, add object points, image points (after refining them)
    if ret:
        objpoints.append(objp)
```

```
        corners2 = cv2.cornerSubPix(gray, corners, (11, 11), (-1, -1),
criteria=(cv2.TERM_CRITERIA_EPS + cv2.TERM_CRITERIA_MAX_ITER, 30, 0.001))
        imgpoints.append(corners2)

        # Draw and display the corners
        img = cv2.drawChessboardCorners(img, (num_corners_x, num_corners_y), corners2, ret)
        cv2.imshow('img', img)
        cv2.waitKey(500)

cv2.destroyAllWindows()

# Perform camera calibration
ret, mtx, dist, rvecs, tvecs = cv2.calibrateCamera(objpoints, imgpoints, gray.shape[::-1], None, None)

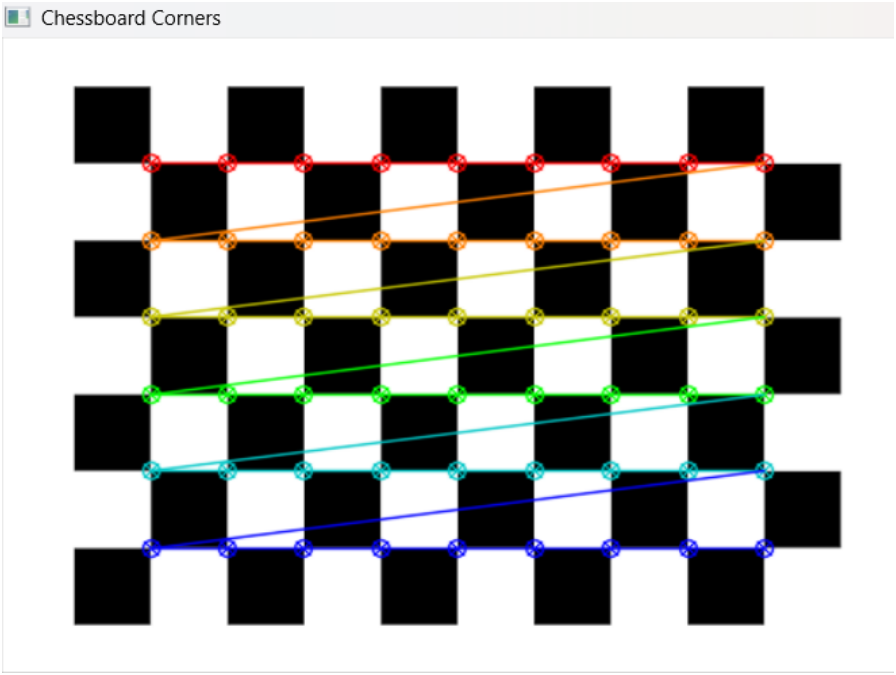
# Save calibration results
np.savez('calibration.npz', mtx=mtx, dist=dist, rvecs=rvecs, tvecs=tvecs)

# Print calibration results
print("Camera matrix:")
print(mtx)
print("\nDistortion coefficients:")
print(dist)
```

output:

```
Camera matrix:
[[5.94061471e+03 0.00000000e+00 5.97367767e+01]
 [0.00000000e+00 5.94900509e+03 1.91644776e+02]
 [0.00000000e+00 0.00000000e+00 1.00000000e+00]]

Distortion coefficients:
[[-1.16584984e+00  4.70099839e+02 -5.40975524e-04 -9.51748088e-04
  -6.22157509e+04]]
```


Chessboard Corners